

6 Limitations

While UMA represents a significant step forward, there are still limitations and areas for improvement. One area to mention is long-range interactions. Our small and medium models use a standard MLIP cutoff distance of 6Å, the actual receptive field is much larger due to message passing, but this can present a problem for inputs where molecules are separated by more than 6Å. For example, if an adsorbate starts at 7Å from a catalyst’s surface the model views these as two independent non-interacting structures. Improvements may also be made in how charge and spin are incorporated into the model [17]. Currently, each discrete charge or spin uses a separate embedding, which limits its ability to generalize to unseen spins or charges.

Our work aims to accelerate scientific discovery in chemistry and materials science using MLIPs; however, potential misuses exist, and we train on datasets designed for applications beneficial to society to mitigate these risks.

7 Discussion and Conclusion

We explore techniques for training MLIPs across diverse DFT tasks using nearly 500 million training examples gathered over numerous datasets. By taking advantage of scaling relationships between dataset size, model capacity and loss, we are able to train models that are competitive with—or better than—task specialized models. Naive approaches to scaling model parameters would result in an accurate but slow model. To address this, we introduced Mixture of Linear Experts, a method to increase model capacity while maintaining inference efficiency. Among our family of models, UMA-S offers a favorable balance between speed and accuracy, capable of performing MD simulations of 1,000 atoms at 1.4ns per day on a single 80GB GPU.

We evaluated UMA across a wide-range of benchmarks for materials, molecules, catalysts, molecular crystals, and metal organic frameworks, demonstrating strong performance across all. For well established benchmarks, such as AdsorbML and Matbench Discovery, we set new state-of-the-art performances. Looking ahead, the development of more challenging benchmarks will be crucial for driving progress in the field.

Despite limitations, our findings suggest that a single model can achieve sufficient accuracy for practical research applications across a broad spectrum of chemistry and materials science applications. This paves the way for universal MLIPs and opens new opportunities for atomic simulations.

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